Airframe Technology Master

MATERIALS & STR.TEST
Lecture 03
Introduction (II) to composite materials

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Introduction

Design of structures made of composite materials:

- Material and the structure are designed concurrently:
  - Designer can vary structural parameters,
  - at the same time vary the material properties

To take advantage of the design flexibility composites offer, it is necessary to understand

- Material selection,
- Manufacturing,
- Material behaviour,
- and Structural Analysis

Introduction

Material selection

Reinforcing fibres	Stiffness/strength (E, σ), Density (specific properties matter for weight-critical parts), Cost and supply-chain maturity	Carbon ⇒ highest modulus & fatigue life, but costly; glass ⇒ good cost/performance; aramid ⇒ exceptional toughness; basalt, natural fibres ⇒ sustainability s.
Matrix material	Service temperature, Toughness & fatigue resistance, Processing temperature window	Thermosets (epoxy, vinyl-ester): low-viscosity, easy impregnation, but not recyclable. Thermoplastics (PEEK, PEKK, PPS): fast cycle times, recyclability, but higher viscosities demand higher pressure or pre-impregnated tapes.
Fibre volume fraction & architecture	Target 55–65 % Vf for high-performance laminates; below ~45 % properties drop off; above ~70 % wet-out becomes difficult. (Derek Hull. (1981).An Introduction to Composite Materials.)	Pick architecture to match loads: unidirectional (UD) for axial loads, ±45° for shear, quasi-isotropic stacks for multiaxial fields, woven or non-crimp fabrics to ease handling.

Manufacturing

Manufacturing process	Typical part size & quality	Cycle-time / cost notes	What to watch
Hand lay-up / wet lay-up	Large, low-pressure parts (boats, wind blades)	Lowest cap-ex, longest cycles	Resin-rich zones & voids; skill heavy
Prepreg + autoclave	Aerospace-grade, <1 % voids	High part quality, high capital & energy	Autoclave size limits; long cure
Out-of-autoclave (OOA) prepreg, VARTM, RTM	Aircraft secondary, ground vehicles	Autoclave-level quality without autoclave; shorter cycle	Vacuum integrity and resin flow control are critical.
Filament winding / pultrusion	Cylindrical or prismatic parts	Near-net-shape, continuous process	Mandrel removal, limited geometries
Additive (CF-PEEK FDM, UV-cured CF-RGUV)	Complex small parts or tooling	Rapid iteration	Porosity and fibre continuity

Material behaviour

- Anisotropy
  - Laminate stiffness and strength vary strongly with fibre orientation; classical lamination theory (CLT) turns ply data into laminate [A B D] matrices
- Key failure modes
  - Matrix cracking (transverse & shear) – Usually the first observable damage, affected by matrix toughness and off-axis plies.
  - Fibre breakage – Governs ultimate strength in 0° plies; brittle for carbon, more graceful for glass.Inter-facial debonding & fibre pull-out – Prominent in short-fibre or natural-fibre systems.
  - Delamination – Driven by out-of-plane stresses; mitigated by ply-drop management, toughened resins and Z-pins. en.wikipedia.orgsciencedirect.com
- Environmental effects – Moisture & temperature (hot-wet) reduce matrix-dominated properties; UV can embrittle exposed resins.
- Fatigue & impact – Carbon/epoxy often retains >60 % static strength at 10^6 cycles; glass/epoxy lower. Low-velocity impacts cause barely visible impact damage (BVID) that must be sized via damage-tolerance criteria.

Structural Analysis

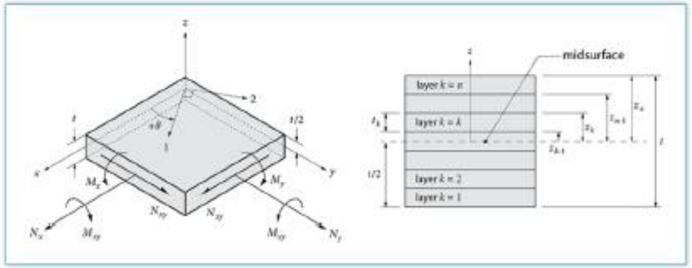
"Hand" calculation or/and Modelling

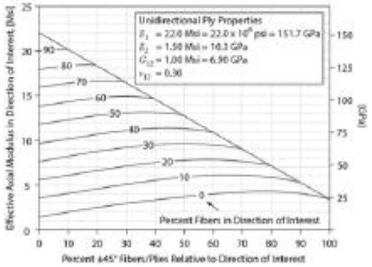
PRELIMINARY DESIGN

'Simple' software

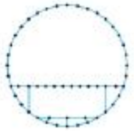
Composite Material are "complicated",

this step can be crucial.





STRUCTURAL FEA

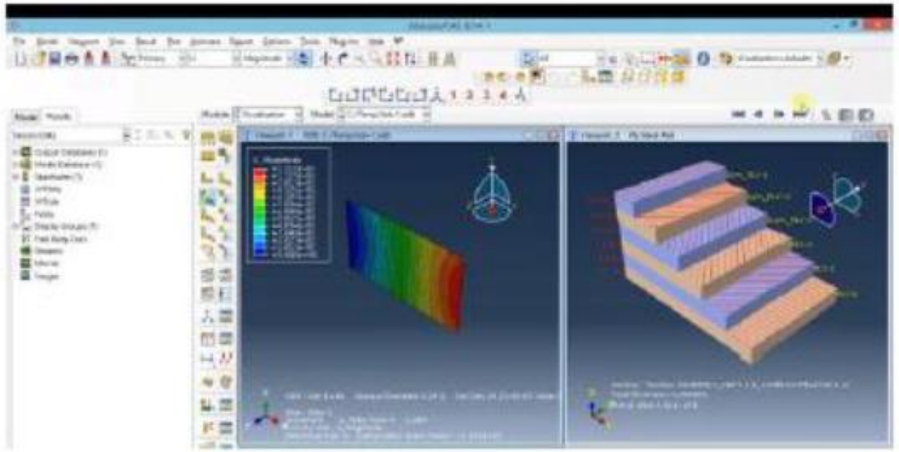


DETAILED ANALYSIS

— Other software

- Abaqus
- LS-Dyna
- …

In general, Finite-Element Analysis



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Getting to know the tools

ESP Composites

…provide software for structural analysis applications.

- Though specifically designed for composites, many of the software modules are useful for metals as well.

All software has been updated to be consistent with the book:

https://bibliotecas.uc3m.es/permalink/f/1nggclj/34UC3M_ALMA21307600470004213

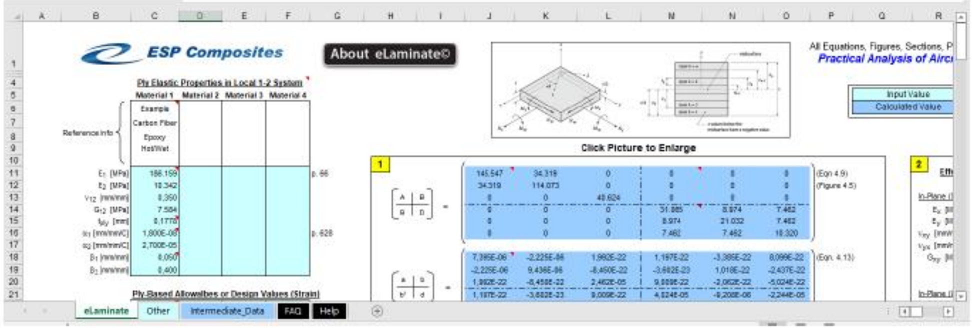
All modules use Microsoft Excel and are compatible with versions 2003+ (32-bit and 64-bit Office).

Free Software at ESP Composites

- eLaminate©, that uses Classical Laminate Theory. Elastic properties such as the the [A], [B], and [D] matrices are calculated. Ply based and laminate based failure criteria are implemented. Mechanical and hygrothermal loads are considered.
- ePlots©, calculates the effective axial modulus, Poisson's ratio, shear modulus, and stress concentration for a symmetric and balanced laminate.
- eSection©, calculates the effective "EA" and "EI" for cross sections that consist of composite elements.
- ePattern©, calculates the individual fastener loads in a 2D pattern. For composite applications, this method may be valid if the load eccentricity is relatively large.
- eColumn©, calculates the critical buckling load for a composite column and is based on the Johnson Euler solution.

https://www.espcomposites.com/software/download.html

eLaminate©



Some comments

Pag. 66 and 628

E_1 = elastic modulus of a ply, longitudinal direction

E_2 = elastic modulus of a ply, transverse direction

G_{12} = shear modulus of a ply

ν_{12} = Poisson's ratio of a ply (the first subscript denotes the loading direction; the second subscript denotes the strain direction)

$\varepsilon_1^{HT}, \varepsilon_2^{HT}, \gamma_{12}^{HT}$ = free strains for a ply that are due to hygrothermal loads (1-2 system)

α_1, α_2 = coefficients of thermal expansion (CTE) in the principal directions of a ply

β_1, β_2 = coefficients of moisture expansion (CME) in the principal directions of a ply; the CME is also known as the coefficient of hygroscopic expansion (CHE)

ΔT = temperature gradient

Δc = moisture gradient

eLaminate©

Some comments

Pag. 169 and Figure 9.5

e_1^{tu} = ultimate tensile strain (strain allowable), longitudinal direction

e_2^{tu} = ultimate tensile strain, transverse direction

e_1^{cu} = ultimate compressive strain, longitudinal direction (negative value)

e_2^{cu} = ultimate compressive strain, transverse direction (negative value)

e_{12}^{s} = in-plane shear strain allowable

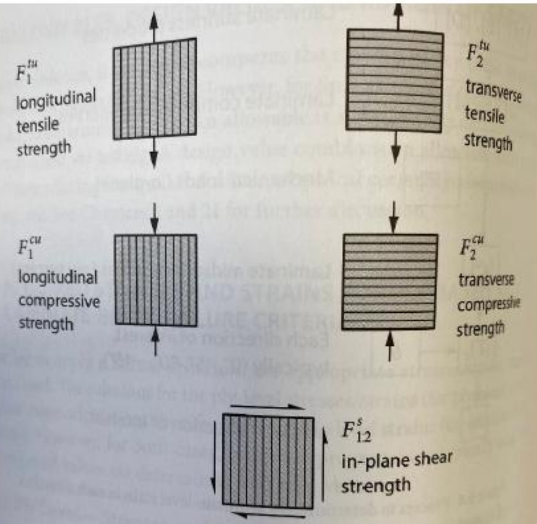


Figure 9.5 Ply strengths (ply stress allowables) for the 5 loading directions

F_1^{tu} = ultimate tensile strength (stress allowable), longitudinal direction

F_2^{tu} = ultimate tensile strength, transverse direction

F_1^{cu} = ultimate compressive strength, longitudinal direction (negative value)

F_2^{cu} = ultimate compressive strength, transverse direction (negative value)

F_{12}^{s} = in-plane shear strength

eLaminate©

Some comments

Eq. (9.9), Eq. (10.6)

and Eq (B.3)

$$MS = \begin{cases} \dfrac{e_{UNT}}{\varepsilon_\theta} - 1 & \text{when} \quad \varepsilon_\theta > 0 \\ \dfrac{e_{UNC}}{\varepsilon_\theta} - 1 & \text{when} \quad \varepsilon_\theta < 0 \end{cases} \tag{9.9}$$

e_{UNT} = preliminary uniaxial tension strain allowable of unnotched laminate

e_{UNC} = preliminary uniaxial compression strain allowable of unnotched laminate

$$\begin{bmatrix} N_x^{HT} \\ N_y^{HT} \\ N_{xy}^{HT} \end{bmatrix} = \sum_{k=1}^{n} [\bar{Q}]_k \begin{bmatrix} \varepsilon_x^{HT} \\ \varepsilon_y^{HT} \\ \gamma_{xy}^{HT} \end{bmatrix}_k \cdot t_k$$

$$\begin{bmatrix} M_x^{HT} \\ M_y^{HT} \\ M_{xy}^{HT} \end{bmatrix} = \sum_{k=1}^{n} [\bar{Q}]_k \begin{bmatrix} \varepsilon_x^{HT} \\ \varepsilon_y^{HT} \\ \gamma_{xy}^{HT} \end{bmatrix}_k \cdot \bar{z}_k t_k \tag{B.3}$$

$$\bar{z}_k = \frac{z_k + z_{k-1}}{2}$$

t_k = ply thickness

$$MS = \begin{cases} \dfrac{e_{OHT}}{\varepsilon_\theta} - 1 & \text{when} \quad \varepsilon_\theta > 0 \\ \dfrac{e_{OHC}}{\varepsilon_\theta} - 1 & \text{when} \quad \varepsilon_\theta < 0 \end{cases} \tag{10.6}$$

ε_θ = remote strain in given orientation (See Chapter 9)

e_{OHT} = open hole tension strain allowable (See Chapter 21)

e_{OHC} = open hole compression strain allowable (See Chapter 21)

eLaminate©

Some comments

Eq. (4.9), Fig. 4,5 and

Eq. (4.13)

$$[A] = \sum_{k=1}^{n} [\bar{Q}]_k (t_k)$$

$$[B] = \sum_{k=1}^{n} [\bar{Q}]_k (t_k \bar{z}_k)$$

$$[D] = \sum_{k=1}^{n} [\bar{Q}]_k \left(\frac{t_k^3}{12} + t_k \bar{z}_k^2 \right)$$

$$\left[\begin{array}{c|c} A & B \\ \hline B & D \end{array}\right]^{-1} = \left[\begin{array}{c|c} a & b \\ \hline b^{T} & d \end{array}\right]$$

$$\begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \\ \hline \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix} = \left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{16} & b_{11} & b_{12} & b_{16} \\ & a_{22} & a_{26} & b_{21} & b_{22} & b_{26} \\ (sym) & & a_{66} & b_{61} & b_{62} & b_{66} \\ \hline b_{11} & b_{21} & b_{61} & d_{11} & d_{12} & d_{16} \\ b_{12} & b_{22} & b_{62} & & d_{22} & d_{26} \\ b_{16} & b_{26} & b_{66} & (sym) & & d_{66} \end{array}\right] \begin{bmatrix} N_x \\ N_y \\ N_{xy} \\ \hline M_x \\ M_y \\ M_{xy} \end{bmatrix}$$

$$\begin{bmatrix} \varepsilon^o \\ \hline \kappa \end{bmatrix} = \left[\begin{array}{c|c} a & b \\ \hline b^{T} & d \end{array}\right] \begin{bmatrix} N \\ \hline M \end{bmatrix}$$

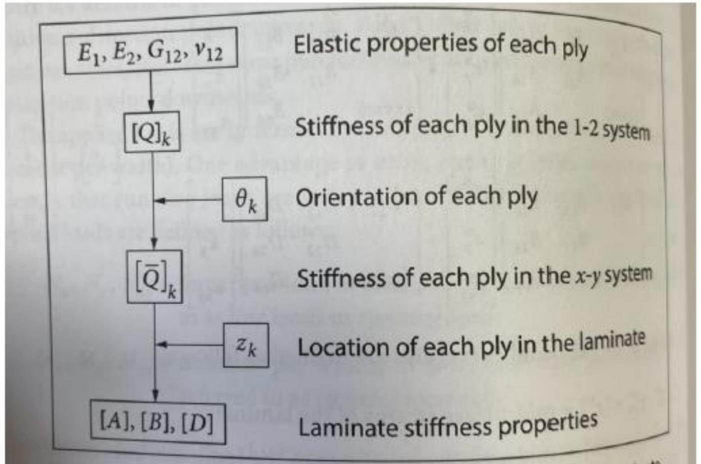


Figure 4.5 Procedure to determine the stiffness properties of a laminate

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Some comments

Eq. (5.6), Eq. (5.12), Eq. (5.17), and Eq (5.18)

$$E_x = \frac{(N_x/t)}{\varepsilon_x^o} = \frac{1}{t \cdot a_{11}} \qquad \nu_{xy} = -\frac{a_{12}}{a_{11}}$$

$$E_y = \frac{1}{t \cdot a_{22}} \qquad \nu_{yx} = -\frac{a_{12}}{a_{22}}$$

$$G_{xy} = \frac{1}{t \cdot a_{66}}$$

$$E_x = \frac{(N_x/t)}{\varepsilon_x^o} = \frac{1}{t \cdot A_{11}^*} \qquad \nu_{xy} = -\frac{A_{12}^*}{A_{11}^*}$$

$$E_y = \frac{1}{t \cdot A_{22}^*} \qquad \nu_{yx} = -\frac{A_{12}^*}{A_{22}^*}$$

$$G_{xy} = \frac{1}{t \cdot A_{66}^*}$$

$$E_x = \frac{1}{t\left(a_{11} + 2 \cdot \xi \cdot b_{11} + \xi^2 d_{11}\right)} = \frac{d_{11}}{t\left(a_{11}d_{11} - b_{11}^2\right)}$$

$$E_y = \frac{d_{22}}{t\left(a_{22}d_{22} - b_{22}^2\right)}$$

$$G_{xy} = \frac{d_{66}}{t\left(a_{66}d_{66} - b_{66}^2\right)}$$

eLaminate©

Some comments

Eq. (5.22), Eq. (B.9) and Eq. (B.10)

$$E_{xb} = \frac{12}{d_{11} \cdot t^3}$$

$$E_{yb} = \frac{12}{d_{22} \cdot t^3}$$

$$\alpha_x = \frac{1}{\Delta T}\left[a_{11} N_x^{HT} + a_{12} N_y^{HT}\right]$$

$$\alpha_y = \frac{1}{\Delta T}\left[a_{12} N_x^{HT} + a_{22} N_y^{HT}\right] \tag{B.9}$$

$$\beta_x = \frac{1}{\Delta c}\left[a_{11} N_x^{HT} + a_{12} N_y^{HT}\right]$$

$$\beta_y = \frac{1}{\Delta c}\left[a_{12} N_x^{HT} + a_{22} N_y^{HT}\right] \tag{B.10}$$

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Some comments

Fig. 7.5

Fig. B.2

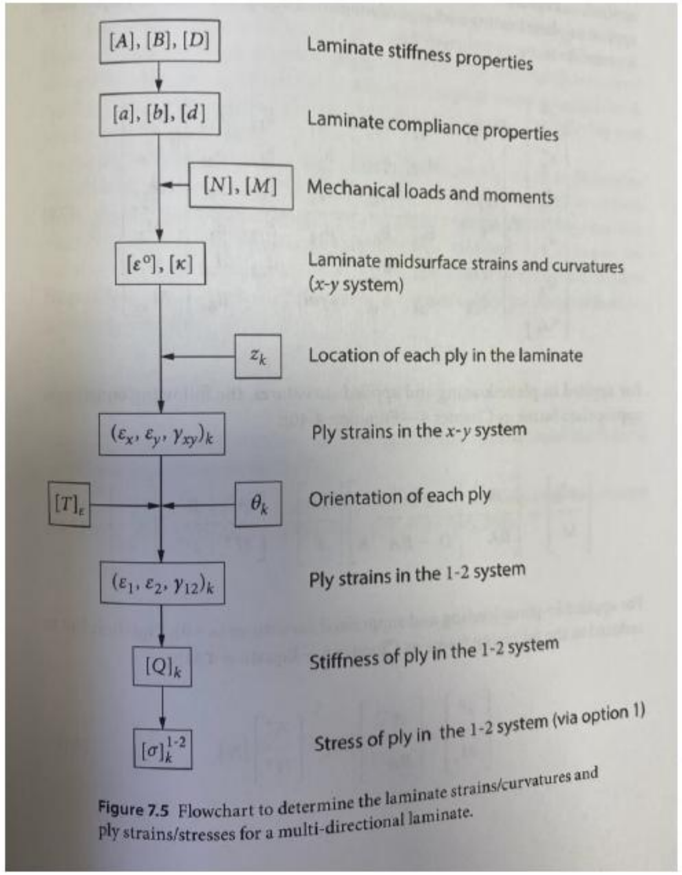

Figure 7.5 Flowchart to determine the laminate strains/curvatures and ply strains/stresses for a multi-directional laminate.

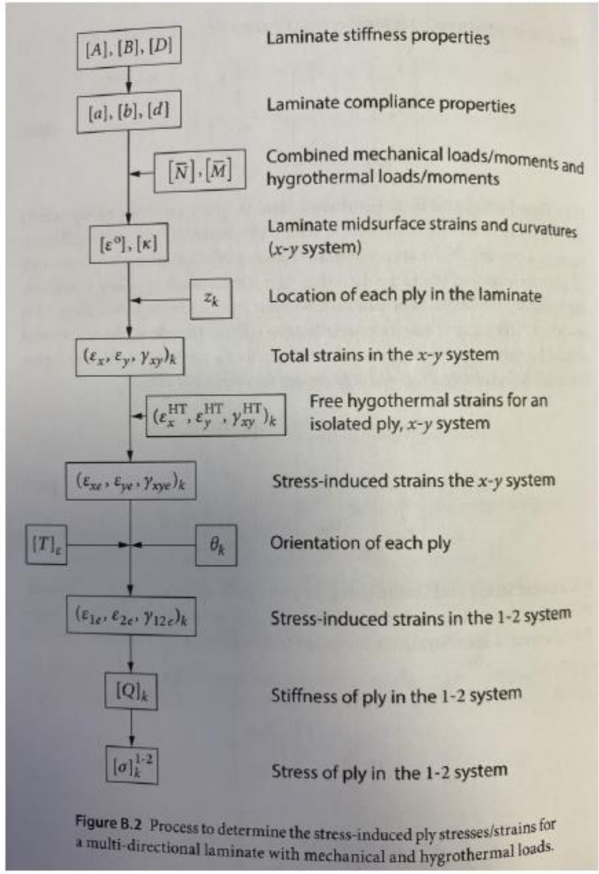

Figure B.2 Process to determine the stress-induced ply stresses/strains for a multi-directional laminate with mechanical and hygrothermal loads.

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Example

Determine the stresses and strains of all the plies that are part ofa cross-ply laminate[0/90]s, which is subjected to an axial force Nx= 100 kN/
m

Data:

E1=160 GPa

E2= 8 Gpa

G12 = 4,5 Gpa

V12 = 0,3

Ply thickness = 0,2 mm

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References

Practical analysis of aircraft composites
Brian.
Esp
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https://www.espcomposites.com/index.html